LAB 3

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Solutions

E1: Individual

Problem

Consider the following MIPS code fragments, each containing two instructions. For each code fragment identify the type of hazard that exists between the two instructions and the registers involved.

a. LD R1, 0(R2)

   DADD R3, R1, R2

b. MULT R1, R2, R3

   DADD R1, R2, R3

c. MULT R1, R2, R3

   MULT R4, R5, R6

d. DADD R1, R2, R3

   SD 2000(R0), R1

e. DADD R1, R2, R3

   SD 2000(R1), R4

Answer

a. **Read After Write (RAW):** The add instruction needs the value stored in R1 by the earlier ld instruction.

b. **Write After Write (WAW**): Both add and mul write to R1, which can cause a conflict if not handled properly.

c. **Structural Hazard:** There's a potential hardware conflict because the multiplier unit might already be in use.

d. **RAW Dependency**: The sd instruction, during its memory access (MEM) stage, depends on the value of R1 calculated by the add instruction.

e**. RAW Dependency**: The sd instruction also relies on R1 in its ALU (execute) stage, which is produced by add.

E2: groups of 2

Problem

a. Explain the behaviour of a 2-bit saturating counter branch predictor.

Show the state of the predictor and the transition for each outcome of the

branch.

b. Consider the following code:

```
for (i=0; i<N; i++)
   if (x[i] == 0)
       y[i] = 0.0;
else
      y[i] = y[i]/x[i];
```

Assume that the assembly code generated is then:

```
loop: L.D F1, 0(R2)
         L.D F2, 0(R3)
         BNEZ F1, else
          ADD.D F2, F0, F0
         BEZ R0, fall
else: DIV.D F2, F2, F1
fall:   DADDI R2, R2, 8
        DADDI R3, R3, 8
        DSUBI R1, R1, 1
        S.D -8(R3), F2
        BNEZ R1, loop
```

where:

- the value of N is already stored in R1
- the base addresses for x and y are stored in R2 and R3, respectively
- register F0 contains the value 0
- register R0 (always) contains the value 0

Assuming that every other element of x has the value 0, starting with the first one, show the outcomes of predictions when a 2-bit saturating counter is used to predict the inner branch BNEZ F1, else. Assume that the initial value of the counter is 00

Answer

a. Branch predictor using a 2-bit saturating counter

Current counter value	Prediction	Actual outcome	New counter value
00	NT	NT	00
00	NT	T	01
01	NT	NT	00
01	NT	T	10
10	T	NT	01
10	T	T	11
11	T	NT	10
11	T	T	11

b. 2-bit counter prediction rate

Iteration	Current counter value	Prediction	Actual outcome	New counter value
1	00	NT	NT	00 (hit)
2	00	NT	T	01 (miss)
3	01	NT	NT	00 (hit)
4	01	NT	T	10 (miss)
5	10	T	NT	01 (miss)
6	10	T	T	11 (hit)
7	11	T	NT	10 (miss)
8	11	T	T	11 (hit)